Measuring Devices and Scales

Notes

Different Scientific Apparatus and its Applications

Apparatus	Application
Altimeter	An instrument used to measure the altitude of an object, mainly in an aeroplane.
Ammeter	An instrument used to measure the electric current in a circuit.
Anemometer	A device used for measuring wind speed and wind pressure.
Actiometer	A device used to measure the heating power or intensity of solar radiation.
Accelerometer	A device that senses the different types of accelerations or vibrations.
Atmometer	A device used for measuring the rate of water evaporation.
Audiometer	An instrument used for measuring or evaluating hearing acuity.
Barograph	A type of Barometer that records the atmospheric pressure over time in graphical form.
Barometer	A device used for measuring atmospheric pressure.
Bolometer	A device to measure the power of incident electromagnetic radiation via the heating of a material with a temperature dependent electrical resistance.
Calipers	A caliper is a device used to measure the distance between two opposite sides of an object.
Calorimeter	A calorimeter is an object used for calorimetry, or the process of measuring the heat of chemical reactions or physical changes as well as heat capacity.
Cardiograph	An instrument for recording graphically the movements of the heart.
Chronometer	It is a specific type of mechanical timepiece tested and certified to meet certain precision standards. It is used by navigators in the sea.
Colorimeter	A device used to measure the absorbance of particular wavelengths of light by a specific solution.
Cathetometer	An instrument to measure vertical distance.
Cryometer	A type of thermometer used to measure the very low temperature of an object.
Cyclotron	An apparatus that accelerates charged particles outwards from the centre along a spiral path.
Crescograph	A device used to measure the growth in plants.
Dilatometer	An instrument that measures volume changes caused by a physical or chemical process.
Dip Circle	A device used to measure the angle between the horizon and the Earth's magnetic field (the dip angle).
Dynamo	An instrument which converts mechanical energy into electrical energy.
Dynamometer	A device used for measuring force, Torque or power of the engine.
Electroencephalograph (EEG)	An instrument used for recording the electrical activity of the brain.
Electrometer	An electrical instrument for measuring electric charge or electrical potential difference.
Electroscope	An instrument used to detect the presence and magnitude of electric charge on a body.
Evaporimeter	An instrument used for measuring the rate of water evaporation from a wet surface to the atmosphere.
Endoscope	An instrument used in medicine to look inside the body.
Fathometer	An apparatus to measure the depth of the sea.
Fluxmeter	An instrument used to measure the magnetic flux.
Galvanometer	An electrical instrument for detecting and indicating electrical current.
Gravimeter	An instrument used for measuring gravitational acceleration.
Gyroscope	A device used for measuring angular velocity.
Hydrometer	An instrument for measuring the relative density of liquids.
Hydrophone	An instrument used for recording or listening to under-water sound.
Hygrometer	An instrument for measuring the relative humidity of the atmosphere.
Hygroscope	An instrument which gives an indication of the humidity of air.

Hypsometer	A device for calibrating thermometers at the boiling point of water at a known height above sea level or for estimating height above sea level by the finding temperature at which water boils.
Kymograph	A device to record muscular motion, changes in blood pressure and other physiological phenomena.
Lactometer	An instrument used to check the purity of milk by measuring its density.
Luxmeter	An equipment that measures the brightness of light falling on an object at a particular area.
Lysimeter	A device used to measure the amount of actual evapotranspiration which is released by the plant.
Magnetometer	It is also known as Magnetic sensor. It is an instrument that measures magnetism-either by magnetization of a magnetic material like a ferromagnet or the direction, strength or relative change of a magnetic field at a particular location.
Manometer	A device to measure pressure of liquids or gases.
Microphone	Its nickname is mic or mike. It is a transducer that converts sound waves into electrical signals.
Microtome	It is a tool used for cutting extremely thin slices of materials, known as sections. Used especially in Biology to observe the sections under a microscope or electron microscope.
Nephoscope	An instrument for measuring the altitude, direction and velocity of clouds.
Odometer	An instrument for measuring the distance travelled by a wheeled vehicle.
Ondometer	An instrument for measuring the frequency/ wavelength of electromagnetic waves.
Periscope	It is an instrument for observation over, around or through an object or condition that prevents direct line of sight observation from the observer's current position.
Phonograph	A device for mechanical recording & reproduction of sound.
Photometer	An instrument for measuring the intensity of light.
Potentiometer	An instrument for measuring voltage by comparison of an unknown voltage with a known reference voltage or to measure the electromotive force (emf) – a differential potential that tends to give rise to an electric current.
Pycnometer	A laboratory device used for measuring the density or specific gravity of materials (liquids or solids).
Pyrheliometer	It is an instrument for the measurement of direct beam solar irradiance.
Pyrometer	A type of remote-sensing radiation thermometer used to measure the high temperature of the surface.
Psychrometer	An apparatus used to measure the relative humidity of the atmosphere.
Rain Gauge	(Udometer, Pluviometer or Ombrometer) It gathers and measures the amount of liquid precipitation over a set of period.
Radiometer	It is a device for measuring the radiant flux of electromagnetic radiation.
Refractometer	A device used for the measurement of an index of refraction.
Salinometer	An instrument used for measuring the salinity of water.
Saccharimeter	An instrument used for measuring the concentration of sugar solutions.
Sextant	An instrument which measures the angular distance between two visible objects. It is used to know the height of celestial bodies.
Sphygmo- manometer	An apparatus used to measure the blood pressure.
Stethoscope	A medical device used for listening to the sounds of the heart.
Speedometer	It measures & displays the instantaneous speed of a vehicle.
Tachometer	An instrument used for measuring rotation of the speed of a shaft or device. It is widely used in automobiles & planes etc.
Viscometer	An instrument used to measure the viscosity of a fluid.
Wind vane	An instrument for showing the direction of the wind.
SONAR	(Sound Navigation and Ranging) – A technique that uses sound propagation (usually underwater, as in submarine navigation) to navigate, communicate with or detect objects on or under the surface of the water, such as other vessels. It uses the echo principle by sending out sound waves.
RADAR	(Radio Detection and Ranging) – Radar is an object - detection system that uses radio waves to determine the range, angle or velocity of objects.

Question Bank

1. Which one of the following instrument is used for locating submerged objects in an ocean ?

(a) Audiometer (b) Galvanometer
(c) Sextant (d) SONAR

U.P. R.O./A.R.O. (Pre) (Re. Exam) 2016
U.P.P.C.S. (Pre) 2000

Ans. (d)

SONAR [Sound Navigation and Ranging] is used for locating submerged objects in an ocean. It is based on a very simple principle i.e. pulse of ultrasonic waves is sent into the water, it strikes the target and bounced back towards the source. It helps to detect or locate submerged submarines and icebergs.

2. SONAR is mostly used by–

(a) Astronauts (b) Doctors
(c) Engineers (d) Navigators

Chhattisgarh P.C.S. (Pre) 2011

Ans. (d)

SONAR is mostly used by Navigators.

3. In SONAR, we use –

(a) Radio waves (b) Audible sound waves
(c) Ultrasonic waves (d) Infrasonic waves

U.P.P.C.S. (Mains) 2013

Ans. (c)

See the explanation of above question.

4. The apparatus used to measure intensity of sound is :

(a) Chronometer (b) Anemometer
(c) Audiophone (d) Audiometer

M.P.P.C.S. (Pre) 1990

Ans. (d)

An Audiometer is used to measure the intensity of sound, while Anemometer is used for measuring wind speed and wind pressure. Chronometer is a timepiece or timing device with a special mechanism for ensuring and adjusting it's accuracy, for use in determining longitude at sea or for any purpose where very exact measurement of time is required. Audiophone is a type of small hearing aid.

5. Which of the following is measured by 'Anemometer' ?

(a) Velocity of water-flow (b) Depth of water
(c) Force of the wind (d) Intensity of light

M.P.P.C.S. (Pre) 2012

Ans. (c)

See the explanation of above question.

6. The velocity of wind is measured by :

(a) Barometer (b) Anemometer
(c) Hydrometer (d) Wind vane

U.P.P.C.S. (Pre) 2016

Ans. (b)

See the explanation of above question.

7. Wind velocity is measured by :

(a) Barometer
(b) Hygrometer
(c) Maximum-Minimum Thermometer
(d) Anemometer

M.P. P.C.S. (Pre) 2022

Ans. (d)

See the explanation of above question.

8. Which one of the following is not correctly matched –

(a) Anemometer – Wind speed
(b) Ammeter – Electric current
(c) Tacheometer – Pressure difference
(d) Pyrometer – High temperature

U.P.P.C.S. (Pre) 1997

Ans. (c)

A tacheometer is a type of theodolite used for rapid measurements and determines, electronically or electro-optically, the distance to target and is highly automated in its operations. The remaining pairs are correctly matched. Pyrometer is a device used for measuring relatively high temperature, such as is encountered in furnaces. Anemometer is a device used for measuring wind speed. An ammeter is a measuring instrument used to measure electric current in a circuit. Electric current is measured in ampere (A).

9. The device to measure electric current is –

(a) Voltmeter (b) Ammeter
(c) Voltameter (d) Potentiometer
(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019
64th B.P.S.C. (Pre) 2018

Ans. (b)

See the explanation of above question.

10. Which of the following is not correctly matched?

(a) Voltmeter-Potential difference
(b) Ammeter-Electric current
(c) Potentiometer-Electromotive force
(d) Galvanometer-Electric resistance
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2022

Ans. (d)

Galvanometer is an instrument for measuring a small electrical current or a function of the current by deflection of a moving coil. The deflection is a mechanical rotation derived from forces resulting from the current. Hence, option (d) is not correctly matched. Rest are correctly matched.

11. Pyrometer is used to measure –

(a) Air pressure (b) Humidity
(c) High temperature (d) Density

Chhattisgarh P.C.S. (Pre) 2011
U.P.P.C.S. (Mains) 2008
Uttarakhand P.C.S. (Pre) 2006
U.P.P.C.S. (Spl.) (Mains) 2004
U.P.P.C.S. (Pre) 1998

Ans. (c)

A Pyrometer is a type of remote-sensing radiation thermometer which is used to measure high temperature. It is based on Stefan-Boltzmann law, which describes that the total radiation emitted by a black body is proportional to the fourth power of its absolute temperature ($E\mu T^4$). It is also used to measure the temperature of distant objects such as sun.

12. Which one of the following devices is used to measure extremely high temperature?

(a) Pyrometer (b) Photometer
(c) Phonometer (d) Pycnometer

U.P. P.C.S. (Mains) 2016

Ans. (a)

See the explanation of above question.

13. Which one of the following thermometers is known as pyrometer?

(a) Thermo-electric thermometers
(b) Radiation thermometers
(c) Gas thermometers
(d) Liquid thermometers

U.P. P.C.S. (Pre) 2016

Ans. (b)

See the explanation of above question.

14. The thermometer which used to measure 2000°C temperature is –

(a) Gas thermometer
(b) Mercury thermometer
(c) Total radiation pyrometer
(d) Steam pressure thermometer

U.P.P.C.S. (Pre) 2009

Ans. (c)

A total radiation pyrometer is used to measure very high temperature. It is suited especially for the measurement of moving objects or any surface that cannot be reached or cannot be touched. The temperature is measured by measuring the thermal radiation.

15. Which one of the following can be used to measure temperature above 1500°C ?

(a) Clinical thermometer
(b) Thermoelectric couple thermometer
(c) Platinum resistance thermometer
(d) Pyrometer

U.P. Lower Sub. (Pre) 2013
U.P. U.D.A./L.D.A. (Spl.) (Pre) 2010

Ans. (d)

Pyrometer is used to measure temperature above 1500°C.

16. 'Pyrheliometer' is used for measuring :

(a) Sun spots
(b) Solar radiation
(c) Air temperature
(d) Temperature of plants

U.P.P.C.S. (Pre) (Re. Exam) 2015

Ans. (b)

Mainly there are two types of devices that are used to measure solar radiations, these are : (i) Pyrheliometer, (ii) Pyranometer.

17. What is measured by manometer?

(a) Air pressure
(b) Pressure of gas
(c) Density of liquids
(d) Pressure of oil on the surface

U.P.P.C.S. (Pre) 1990

Ans. (b)

Manometer is an instrument that uses a column of liquid to measure pressure of liquids or gases, commonly referred as pressure measuring instrument.

18. Match List-I with List-II and select the correct answer using the codes given below:

List - I (Instrument)	**List - II (Measuring quantity)**
A. Ammeter	**1. Pressure**
B. Hygrometer	**2. Weight**
C. Spring balance	**3. Electric current**
D. Barometer	**4. Relative humidity**

Code :

	A	B	C	D
(a)	2	3	4	1
(b)	3	4	2	1
(c)	4	1	2	3
(d)	1	2	3	4

U.P.U.D.A./L.D.A (Pre) 2001

Ans. (b)

An ammeter is a measuring instrument used to measure the electric current in a circuit. A hygrometer is used for measuring the moisture content, which is known as humidity in the atmosphere. A spring balance is a weighing apparatus that measures different weights by the tension of a spring. A barometer is used to measure atmospheric pressure.

19. Which one of the following pair is not correctly matched?

(a) Odometer : Measuring instrument for distance covered by motor wheels

(b) Ondometer : Measuring instrument for frequency of electromagnetic waves

(c) Audiometer : Device for measuring sound intensity

(d) Ammeter : Measuring instrument for electric power

U.P. Lower Sub. (Pre) 2015

Ans. (d)

Ammeter is an instrument used to measure the electric current not for measuring electric power. Thus, option (d) is not correctly matched.

20. The meter that is used to measure the distance moved by the vehicle is known as :

(a) Ammeter (b) Speedometer

(c) Chronometer (d) Odometer

70th B.P.S.C. (Pre) 2024

Ans. (d)

An odometer is used to measure the distance travelled by a wheeled vehicle. The device may be electronic, mechanical or a combination of the two (electromechanical).
Ammeter – used to measure the electric current in a circuit
Speedometer – used to measure the speed of a vehicle
Chronometer – an extraordinary accurate mechanical timepiece, particularly for use in maritime navigation to determine longitude

21. Which instrument is used to measure atmospheric pressure?

(a) Hydrometer (b) Barometer

(c) Manometer (d) Hygrometer

U.P.P.S.C. (GIC) 2010
M.P.P.C.S. (Pre) 2000

Ans. (b)

A barometer is a scientific instrument used in meteorology to measure atmospheric pressure. A sudden fall of mercury in a barometer indicates the thunderstorm.

22. The liquid/liquids used in the simple barometer is / are :

(a) Water (b) Mercury

(c) Alcohol (d) All of the above

M.P.P.C.S. (Pre) 2016

Ans. (b)

A barometer is an instrument for measuring atmospheric pressure. There are different types of barometers e.g. water-based barometers, mercury barometers, vaccum pump oil barometer, aneroid (nonliquid) barometers etc. Generally, a simple barometer consists of a long glass tube (closed at one end and open at the other) filled with mercury and turned upside down into a container of mercury.

23. The density of milk can be obtained by the use of :

(a) Hydrometer (b) Butyrometer

(c) Lactometer (d) Thermometer

M.P.P.C.S. (Pre) 2006

Ans. (c)

Lactometer is used for the measurement of the density of milk. Butyrometer is used to measure fat content in milk or milk products.

24. The density of milk is measured by?

(a) Lactometer (b) Hydrometer

(c) Barometer (d) Hygrometer

M.P.P.C.S. (Pre) 2015

Ans. (a)

See the explanation of above question.

25. Relative humidity is measured by :

(a) Hydrometer (b) Hygrometer

(c) Lactometer (d) Potentiometer

U.P.P.C.S. (Pre) 1996
U.P.P.C.S. (Pre) 1995

Ans. (b)

Hygrometer is a device used to determine the relative humidity of the atmosphere. Lactometer is used to measure the density of milk, hydrometer is used for measuring the relative density of liquids while potentiometer is used to measure the electric potential (voltage) in a circuit.

26. Which instrument is used for measuring humidity in the air?

(a) Hydrometer (b) Hygrometer

(c) Spectrometer (d) Eudiometer

Chhattisgarh P.C.S. (Pre) 2008
U.P.P.S.C. (R.I.) 2014

Ans. (b)

A hygrometer is an instrument used to measure the moisture content in the atmosphere which is also called as humidity in the the air. Spectrometer is an instrument used to measure properties of light over a specific portion of the electromagnetic spectrum. A Eudiometer is a laboratory device that measures the change in volume of a gas mixture following a physical or chemical change.

27. Hygrometer is used to measure –
(a) Humidity in atmosphere
(b) Atmospheric pressure
(c) High temperature
(d) Velocity of wind

Uttarakhand Lower Sub. (Pre) 2010

Ans. (a)

See the explanation of above question.

28. Which instrument is used to measure humidity?
(a) Hydrometer (b) Hygrometer
(c) Pyrometer (d) Lactometer
(e) None of the above / More than one of the above

64th B.P.C.S. (Pre) 2018

Ans. (b)

See the explanation of above question.

29. Which of the following device quantify the humidity in Air?
(a) Thermostat (b) Pyrometer
(c) Hypsometer (d) Hygrometer

R.A.S./R.T.S. (Pre) 1996

Ans. (d)

Hygrometer is a device which quantifies the humidity in the air. Pyrometer is used for measuring high temperature and hypsometer is used to measure height or altitude. Thermostat is a device which is used for regulating the temperature of a system so that the system's temperature is maintained near a desired set point temperature.

30. 'Ringelmann Scale' is used to measure the density of the following :
(a) fog (b) noise
(c) polluted water (d) smoke

U.P. R.O./A.R.O. (Mains) 2021

Ans. (d)

The Ringelmann Scale is a scale for measuring the apparent density or opacity of smoke. It was developed by a French professor of agricultural engineering Maximilien Ringelmann, who first specified the scale in 1888. The Ringelmann Smoke Chart fulfills an important need in smoke abatement work and in certain problems in the combustion of fuels.

31. Which of the following is correctly matched?
(a) Thermoresistor – Electronic Thermometer
(b) Capacitor – Thermometer
(c) Bipolar Junction Transistor – Rectifier
(d) Junction Diode – Amplifier

U.P. R.O./A.R.O. (Pre) 2016

Ans. (a)

Thermoresistor is a device which acts as an electronic thermometer. This device changes its resistance with changes in temperature. So, option (a) is correctly matched.

32. Electroencephalogram (EEG) is used in monitoring :
(a) Heart (b) Liver
(c) Pancreas (d) Brain
(e) None of the above / More than one of the above

64th B.P.C.S. (Pre) 2018

Ans. (d)

An EEG is a test that detects electrical activity of the brain using small, metal discs (electrodes) attached to the scalp. Brain cells communicate via electrical impulses and are active all the time, even when a person is sleeping. This activity shows up wavy lines on an EEG recording.

33. Match List-I with List-II and select the correct answer from the codes given below the lists :

List-I	List-II
A. Stethoscope	**1. To measure intensity of light**
B. Sphygmomanometer	**2. To check purity of gold**
C. Caratometer	**3. To hear heart sound**
D. Luxmeter	**4. To measure blood pressure**

Code :

	A	B	C	D
(a)	1	2	3	4
(b)	4	3	2	1
(c)	3	4	2	1
(d)	2	1	4	3

U.P.P.C.S. (Pre) 2008

Ans. (c)

Stethoscope is an acoustic medical device used for listening to the action of someone's heart sound. Sphygmomanometer is used to measure blood pressure. Caratometer is an internationally acclaimed device for checking the purity of gold. Luxmeter is used to measure the intensity of light.

34. The name of the equipment used for measuring blood pressure is :
(a) Tacheometer (b) Sphygmomanometer
(c) Actiometer (d) Barometer

U.P.P.C.S. (Pre) 2007

Ans. (b)

See the explanation of above question.

35. The apparatus used to measure the intensity of light is known as –
(a) Anemometer (b) Colorimeter
(c) Luxmeter (d) Altimeter

U.P.P.C.S. (Spl.) (Mains) 2004

Ans. (c)

See the explanation of above question.

36. Radar is used for :

(a) Detecting objects by using light waves
(b) Reflecting sound waves to detect objects
(c) Determining the presence and location of objects with radio waves
(d) Tracking rain-bearing clouds

U.P.P.C.S. (Pre) 2008
U.P.P.C.S. (Pre) 1996
U.P.U.D.A./L.D.A. (Pre) 2001
U.P. Lower Sub. (Pre) 2004

Ans. (c)

RADAR stands for Radio Detection and Ranging. As indicated by the name, it is based on the use of radio waves. It refers to the technique of using radio waves to detect the presence of objects. Today, it is used for a wide variety of applications, such as to determine the range, angle or velocity of objects. It can be used to detect aircraft, ships, spacecraft, guided missiles, vehicles, weather formations, and terrain.

37. Which of the following quantities is measured on the Richter scale?

(a) Speed of a glacier
(b) Population growth
(c) Intensity of an earthquake
(d) Temperature inside the earth

R.A.S./R.T.S.(Pre) 2003

Ans. (c)

The Richter scale is the most common standard of measurement for an earthquake. It was invented in 1935 by Charles F. Richter of the California Institute of Technology as a mathematical device to compose the size of earthquake.The Richter scale is used to rate the magnitude of an earthquake, that is the amount of energy released during an earthquake. **Note :** It is to be noted that intensity of an earthquake (extent of damage) is rated on the Mercalli scale. However, in Commissions' questions intensity of earthquakes generally refers to the power or magnitude of earthquakes.

38. The intensity of earthquakes is measured :

(a) On the Richter scale (b) On the Kelvin scale
(c) In decibel (d) In Pascal

U.P. Lower Sub. (Pre) 2015
U.P. Lower Sub. (Pre) 2004

Ans. (a)

See the explanation of above question.

39. Richter scale is used for measuring –

(a) Velocity of sound
(b) Intensity of light
(c) Amplitude of seismic waves
(d) Intensity of sound

U.P.P.C.S. (Spl.) (Mains) 2008

Ans. (c)

See the explanation of above question.

40. Which one is not correctly matched ?

(a) Celsius	–	Temperature
(b) Kilowatt hour	–	Electricity
(c) Rh factor	–	Blood
(d) Richter scale	–	Humidity

Uttarakhand P.C.S. (Pre) 2002

Ans. (d)

Richter scale is not used to measure humidity. In fact, it is used to rate the magnitude of an earthquake. Remaining pairs are correctly matched.

41. Which of the following is a seismometer device?

(a) Crescograph (b) Seismograph
(c) Geiger Counter (d) Raingauge

U.P.P.C.S. (Pre) 1990

Ans. (b)

Earthquake generates seismic waves which can be detected with a sensitive instrument called Seismograph. Crescograph is a device for measuring growth in plants. It was invented by Sir Jagadish Chandra Bose in the early 20th century. Raingauge is used to measure rain. Geiger Counter is a 'particle detector' device that detects radioactivity or radiation.

42. Which of the following instruments is used to record seismic waves?

(a) Seismogram (b) Seismograph
(c) Seismoscope (d) Seismometer

U.P. P.C.S. (Mains) 2017

Ans. (b)

See the explanation of above question.

43. Which one of the following devices is used to measure the intensity of earthquake?

(a) Seismograph (b) Stethoscope
(c) Cosmograph (d) Periscope

U.P.P.C.S. (Mains) 2014
M.P.P.C.S. (Pre) 2015

Ans. (a)

See the explanation of above question.

44. What does a 'seismograph' record ?

(a) Heart-beats (b) Atmospheric pressure
(c) Earthquake (d) None of these

M.P.P.C.S. (Pre) 1995

Ans. (c)

See the explanation of above question.

45. Match List-I with List-II and select the correct answer from the codes given below the lists :

List-I	**List-II**
A. Earthquake	**1. Ammeter**
B. Height	**2. Seismograph**
C. Electric current	**3. Altimeter**
D. Resistance	**4. Ohm**

Code :

	A	B	C	D
(a)	2	3	1	4
(b)	2	3	4	1
(c)	1	3	4	2
(d)	3	2	4	1

U.P.P.C.S. (Pre) 1992

Ans. (a)

An altimeter or an altitude meter is an instrument used to measure the altitude of an object above a fixed level. Ammeter is used to measure electric current in a circuit. A seismograph is used to measure earthquakes and Ohm is the unit of electrical resistance.

46. Match the correct :

A. Fathometer	**1. Atmospheric pressure**
B. Barometer	**2. Atmospheric humidity**
C. Hygrometer	**3. Height**
D. Altimeter	**4. Depth of sea**

Code :

	A	B	C	D
(a)	2	3	1	4
(b)	4	1	2	3
(c)	4	2	3	1
(d)	3	1	2	4

Chhattisgarh P.C.S. (Pre) 2003

Ans. (b)

Fathometer is an instrument used to determine the depth of water or a submerged object by means of ultrasonic waves. The barometer is used to measure atmospheric pressure, hygrometer is used to measure atmospheric moisture and altimeter is used to measure altitude/height of an object above a fixed level.

47. Fathometer is used to measure :

(a) Earthquake (b) Rain
(c) Depth of sea (d) Sound intensity

M.P.P.C.S. (Pre) 2015

Ans. (c)

See the explanation of above question.

48. Match List-I with List-II and select the correct answer from the codes given below the lists :

List - I	**List - II**
A. Anemometer	**1. Earthquakes**
B. Seismograph	**2. Atmospheric Pressure**
C. Barograph	**3. Wind velocity**
D. Hygrometer	**4. Humidity**

Code :

	A	B	C	D
(a)	1	2	3	4
(b)	4	1	2	3
(c)	4	1	3	2
(d)	3	1	2	4

U.P.P.C.S. (Mains) 2012
Uttarakhand Lower Sub. (Pre) 2010

Ans. (d)

Anemometer	:	Wind velocity
Seismograph	:	Earthquake
Barograph	:	Atmospheric pressure
Hygrometer	:	Humidity

49. Which one of the following is not correctly matched :

(a) Manometer – Pressure
(b) Carburetor – Internal combustion engine
(c) Cardiograph – Heart movement
(d) Seismometer – Curvature of surface

U.P.P.C.S. (Pre) 1999
U.P.P.S.C. (GIC) 2010

Ans. (d)

Among the given options, option (d) is not correctly matched. Seismometer is an instrument used to measure seismic waves generated by earthquakes. Carburetor is a device that blends air and fuel for an internal combustion engine. Cardiograph is an instrument used to record the mechanical movements of the heart. Manometer is an instrument that uses a column of liquid to measure pressure.

50. Phonometer is used to measure which one of the following?

(a) The power of brightness of light
(b) Extremely high temperature
(c) Frequency of electromagnetic wave
(d) Atmospheric humidity

U.P.P.C.S. (Mains) 2014

Ans. (*)

Phonometer is an instrument which is used for testing the force of the human voice in speaking. Photometer is used for measuring the intensity of light.

51. The apparatus used for detecting lie is known as –

(a) Polygraph (b) Pyrometer
(c) Gyroscope (d) Kymograph

U.P. Lower Sub. (Pre) 2013

Ans. (a)